Google Developer Group
Bandung Institute of Technology

Team name	Tide Ledger
Project title	RapiLPJ: Pre-Submission Audit for Student Organization Activity Reports
Selected case	Case 1: Smarter Campus and Learning
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Abstract

1. Problem. Student treasurers manage campus funds without accounting training under strict bylaws: physical stamps, ledger balance, meal attendance ratios (1 signed list per session), revenue stamps on receipts exceeding Rp5,000,000 under Law 10/2020, PPh 21 taxes, and deadlines. At Brawijaya Computer Science, incomplete reports freeze subsequent funding and peer learning [6]. Treasurers lose valuable study hours to mechanical corrections, turning accountability into administrative anxiety.

2. Intended Users. Primary users: student organization treasurers and secretaries (BEM and clubs) lacking accounting training. Secondary users: faculty student affairs verifiers (Biro Kemahasiswaan and student council auditors) conducting pre-disbursement desk reviews.

3. Supporting Observations or Evidence. Faculty research at Universitas Atma Jaya Makassar documented unstandardized extracurricular documentation and manual record-keeping at the Student Affairs Bureau [1]. Disparate guidelines compound friction: UB mandates stamped receipts and ledger balances [2]; UGM requires Rp10,000 stamps on receipts exceeding Rp5,000,000 under Law 10/2020 and photocopies for thermal slips [3]; ITS enforces 10-day deadlines [4] and tax inclusion [5]. Student officers consistently struggle with tax deductions [6].

4. Proposed Solution and Core User Flow. RapiLPJ transforms administrative triage into experiential fiscal learning in three steps: (1) Intake & Human-Confirmed Audit: Gemini 2.0 Flash extracts fiscal metadata from receipt photos into JSON without arithmetic. Treasurers confirm extracted fields side-by-side with receipts before a client-side engine audits rules (duplicate invoices, dates, >Rp5M stamps, 1:1 attendance, 5% PPh 21, >10% variances), persisting .lpj files. (2) Gated Fiscal Literacy & Addenda: Infraction export is gated until treasurers pass micro-quizzes on locked statutory rationale (Law 10/2020, archival retention); a deterministic template drafts variance addenda. (3) Faculty Desk-Check Portal: Treasurers export an indexed PDF dossier with a SHA-256 hash. Verifiers load the package into a desk-check screen to verify hash integrity, recommending clearance or flagging discrepancies in seconds.

5. How It Differs from Existing Alternatives. Pure LLMs hallucinate calculations on balance sheets. Spreadsheets lack statutory checks and remediation loops. Campus portals merely collect finished files without triage. RapiLPJ enforces a strict neuro-symbolic boundary: Gemini 2.0 Flash is restricted to visual perception and JSON extraction with mandatory human confirmation, while financial arithmetic, compliance verification, and addendum drafting are 100% reserved for transparent, deterministic client-side code.

6. Planned Technologies. A privacy-preserving client-side stack. Perception Layer: Gemini 2.0 Flash Multimodal API (client BYOK) for receipt extraction into JSON. Deterministic Kernel: Open-source JavaScript engine for statutory evaluation, locked addendum templating, FileReader receipt compilation, .lpj persistence, and SHA-256 hashing. Presentation: Accessible HTML5/CSS3 (bilingual EN/ID, light/dark) deployed statically on Vercel CDN with zero persistent database storage, transmitting images solely to the user-authenticated Gemini endpoint.

7. Expected Benefit and Development Plan. Expected benefit: evaluated via three pilot metrics with 30 treasurers and 2 faculty advisors at Universitas Atma Jaya Makassar: preparation time in minutes, mechanical defect counts before vs. after pre-audit, and pre/post fiscal literacy quiz scores. Development plan: Semifinal phase benchmarks single-receipt multimodal perception and the primary UAJM faculty preset; Final Day sprint delivers multi-page document batching and QR-verifiable advisor sign-off. Modular toggles mitigate guideline shifts.

Source Acknowledgments. [1] P. Siada & A. L. Tungadi, TEMATIKA J. Penelit. Tek. Inform. Sist. Inf. UAJM, 2021, doi:10.56963/tematika.v9i1.119 (studi sistem informasi portofolio kemahasiswaan BKAM UAJM). [2] Pedoman LPJ, FIB Universitas Brawijaya, fib.ub.ac.id. [3] Prosedur Proposal & LPJ SIPRO, FEB UGM & UU Bea Meterai No. 10/2020. [4] SOP HIMA, Desain Interior ITS, its.ac.id. [5] SOP LPJ Mahasiswa Biologi, ITS, its.ac.id. [6] Surat Edaran Mekanisme LPJ Ormawa, FILKOM UB, 2026. AI disclosure: Claude and Gemini supported research, deterministic engine code, and drafting.